

Cubic Function Report

Polynomial

$$f(q) = aq^3 + bq^2 + cq + d$$

Naive

$$g_{\text{naive}}(X) = X^3a + X^2b + Xc + d$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 6a\frac{\Delta^2}{\epsilon}q + aq^3 + 2b\frac{\Delta^2}{\epsilon} + bq^2 + cq + d$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) = & 720a^2\frac{\Delta^6}{\epsilon} + 324a^2\frac{\Delta^4}{\epsilon}q^2 + 18a^2\frac{\Delta^2}{\epsilon}q^4 + 216ab\frac{\Delta^4}{\epsilon}q + 24ab\frac{\Delta^2}{\epsilon}q^3 \\ & + 48a\frac{\Delta^4}{\epsilon}c + 12a\frac{\Delta^2}{\epsilon}cq^2 + 20b^2\frac{\Delta^4}{\epsilon} + 8b^2\frac{\Delta^2}{\epsilon}q^2 + 8b\frac{\Delta^2}{\epsilon}cq + 2\frac{\Delta^2}{\epsilon}c^2\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) = & 720a^2\frac{\Delta^6}{\epsilon} + 360a^2\frac{\Delta^4}{\epsilon}q^2 + 18a^2\frac{\Delta^2}{\epsilon}q^4 + 240ab\frac{\Delta^4}{\epsilon}q + 24ab\frac{\Delta^2}{\epsilon}q^3 \\ & + 48a\frac{\Delta^4}{\epsilon}c + 12a\frac{\Delta^2}{\epsilon}cq^2 + 24b^2\frac{\Delta^4}{\epsilon} + 8b^2\frac{\Delta^2}{\epsilon}q^2 + 8b\frac{\Delta^2}{\epsilon}cq + 2\frac{\Delta^2}{\epsilon}c^2\end{aligned}$$

Unbiased

$$g_{\text{unb}}(X) = X^3a + X^2b - 6Xa\frac{\Delta^2}{\epsilon} + Xc - 2b\frac{\Delta^2}{\epsilon} + d$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3 + bq^2 + cq + d$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) = & 504a^2\frac{\Delta^6}{\epsilon} + 252a^2\frac{\Delta^4}{\epsilon}q^2 + 18a^2\frac{\Delta^2}{\epsilon}q^4 + 168ab\frac{\Delta^4}{\epsilon}q + 24ab\frac{\Delta^2}{\epsilon}q^3 \\ & + 24a\frac{\Delta^4}{\epsilon}c + 12a\frac{\Delta^2}{\epsilon}cq^2 + 20b^2\frac{\Delta^4}{\epsilon} + 8b^2\frac{\Delta^2}{\epsilon}q^2 + 8b\frac{\Delta^2}{\epsilon}cq + 2\frac{\Delta^2}{\epsilon}c^2\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{unb}}) = & 504a^2\frac{\Delta^6}{\epsilon} + 252a^2\frac{\Delta^4}{\epsilon}q^2 + 18a^2\frac{\Delta^2}{\epsilon}q^4 + 168ab\frac{\Delta^4}{\epsilon}q + 24ab\frac{\Delta^2}{\epsilon}q^3 \\ & + 24a\frac{\Delta^4}{\epsilon}c + 12a\frac{\Delta^2}{\epsilon}cq^2 + 20b^2\frac{\Delta^4}{\epsilon} + 8b^2\frac{\Delta^2}{\epsilon}q^2 + 8b\frac{\Delta^2}{\epsilon}cq + 2\frac{\Delta^2}{\epsilon}c^2\end{aligned}$$

Comparison

$$\text{mean gap} = -6a\frac{\Delta^2}{\epsilon}q - 2b\frac{\Delta^2}{\epsilon}$$

$$\text{variance gap} = -216a^2\frac{\Delta^6}{\epsilon} - 72a^2\frac{\Delta^4}{\epsilon}q^2 - 48ab\frac{\Delta^4}{\epsilon}q - 24a\frac{\Delta^4}{\epsilon}c$$

$$\text{MSE gap} = -216a^2\frac{\Delta^6}{\epsilon} - 108a^2\frac{\Delta^4}{\epsilon}q^2 - 72ab\frac{\Delta^4}{\epsilon}q - 24a\frac{\Delta^4}{\epsilon}c - 4b^2\frac{\Delta^4}{\epsilon}$$